

Question_3:

DS	Session	Arrays	Question Information	Level 1	Challenge 23
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Problem Description:

Rigesh is an electronic shop owner. Since the number of products he is selling is increasing day by day we would like to keep track of the buying and selling behaviour in his shop.

So given the cost of stock on each day in an array $A[]$ of size N . Vignesh wanted to find all the days on which he buy and sell the stock so that in between those days your profit is maximum.

Constraints:

$1 \leq t \leq 10$

$1 \leq n \leq 10$

Input Format:

First line contains number of test cases T .

First line of each test case contains an integer value N denoting the number of days, followed by an array of stock prices of N days.

Output Format:

For each testcase, output all the days with profit in a single line.

If there is no profit then print "No Profit".

Logical Test Cases

Test Case 1

INPUT (STDIN)

1
5
130 204 150 175 140

EXPECTED OUTPUT

(0 1) (2 3)

Test Case 2

INPUT (STDIN)

2
5
131 124 110 175 110
3
23 45 67

EXPECTED OUTPUT

(2 3)
(0 2)

Mandatory Test Cases

Test Case 1

KEYWORD

if(arr[i]>arr[i-1])

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

17

Test Case 2

TOKEN COUNT

350

Test Case 3

NLOC

71

Code:

```
1  /*
2  Question_3:
3  Find all buy and sell days where the stock price increases
4  and profit can be made. Print "No Profit" if no profit exists.
5  */
6
7  #include <iostream>
8  using namespace std;
9
10 int main()
11 {
12     int T;
13     cin >> T;
14
15     while (T--)
16     {
17         int n;
18         cin >> n;
19
20         int arr[100];
21         for (int i = 0; i < n; i++)
22         {
23             cin >> arr[i];
24         }
25         int buy = -1;
26         bool profit = false;
27         for (int i = 1; i < n; i++)
28         {
29             if(arr[i]>arr[i-1])
30             {
31                 if (buy == -1)
```

```
32         {
33             buy = i - 1;
34         }
35         if (i == n - 1 || arr[i] >= arr[i + 1])
36         {
37             cout << "(" << buy << " " << i << ") ";
38             profit = true;
39             buy = -1;
40         }
41     }
42     else
43     {
44         buy = -1;
45     }
46 }
47 if (!profit)
48 {
49     cout << "No Profit";
50 }
51 cout << endl;
52 }
53 return 0;
54 }
```

Output:

Test Case_1:

```
1
5
130 204 150 175 140
(0 1) (2 3)
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
2
5
131 124 110 175 110
(2 3)
3
23 45 67
(0 2)
PS C:\Users\Prabin Yadav\
```